

Derivative Notation and Applications

David Puelz

The University of Austin

Outline

1. **The $\mathcal{D}_t$ Operator** – Notation and meaning (15 min)
2. **Application: Physics** – Position, velocity, acceleration (15 min)
3. **Application: Economics** – Marginal cost and revenue (15 min)
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1 The $\mathcal{D}_t$ Operator

1.1 Notation and Pronunciation

Definition: We write $\mathcal{D}_t f(t)$ to denote the derivative of f with respect to t .

Pronunciation: “D-sub-t of f ” or “the rate of change of f with respect to t .”

Equivalent notations:

$$\mathcal{D}_t f(t) = f'(t) = \frac{df}{dt} = \frac{d}{dt} f(t)$$

Examples:

$$\mathcal{D}_t(t^2) = 2t$$

$$\mathcal{D}_t(t^5) = 5t^4$$

$$\mathcal{D}_t(7) = 0$$

$$\mathcal{D}_t(3t + 8) = 3$$

1.2 Why Specify “With Respect To”?

The key insight: A quantity may depend on *multiple* variables. The derivative tells us how the quantity changes when we vary *one specific variable*, holding others fixed.

Example: The area of a rectangle with width w and height h :

$$A(w, h) = w \cdot h$$

Two different derivatives:

- $\mathcal{D}_w A = h$ (rate of change of area with respect to width, holding height fixed)
- $\mathcal{D}_h A = w$ (rate of change of area with respect to height, holding width fixed)

These answer different questions!

1.3 The Derivative Transforms Functions

Important concept: The derivative operator $\mathcal{D}_t$ takes a function as input and produces a new function as output.

$$\text{Function } f(t) \xrightarrow{\mathcal{D}_t} \text{New function } f'(t)$$

Example chain:

$$\begin{aligned} f(t) &= t^4 \\ \mathcal{D}_t f(t) &= 4t^3 \\ \mathcal{D}_t(\mathcal{D}_t f(t)) &= \mathcal{D}_t(4t^3) = 12t^2 \end{aligned}$$

Second derivative notation:

$$\mathcal{D}_t^2 f(t) = \mathcal{D}_t(\mathcal{D}_t f(t)) = f''(t) = \frac{d^2 f}{dt^2}$$

2 Application: Physics

2.1 The Kinematic Chain

Position, velocity, and acceleration: For an object moving along a line with position $x(t)$:

$$\begin{aligned} \text{Position: } & x(t) \\ \text{Velocity: } & v(t) = \mathcal{D}_t x(t) = x'(t) \\ \text{Acceleration: } & a(t) = \mathcal{D}_t v(t) = \mathcal{D}_t^2 x(t) = x''(t) \end{aligned}$$

Interpretation:

- Velocity is the rate of change of position
- Acceleration is the rate of change of velocity

2.2 Example: Free Fall

Setup: An object is dropped from height $h_0 = 100$ meters. Its height above ground is:

$$h(t) = 100 - \frac{1}{2}(9.8)t^2 = 100 - 4.9t^2$$

Find velocity and acceleration:

$$v(t) = \mathcal{D}_t h(t) = \mathcal{D}_t(100 - 4.9t^2) = -9.8t$$

$$a(t) = \mathcal{D}_t v(t) = \mathcal{D}_t(-9.8t) = -9.8$$

Interpretation:

- $v(t) = -9.8t$: Velocity is negative (falling down) and increases in magnitude
- $a(t) = -9.8$: Constant acceleration due to gravity

At $t = 2$ seconds:

- Height: $h(2) = 100 - 4.9(4) = 80.4$ m
- Velocity: $v(2) = -9.8(2) = -19.6$ m/s (downward)
- Acceleration: $a(2) = -9.8$ m/s²

2.3 Example: Projectile Motion

Setup: A ball is thrown upward with initial velocity $v_0 = 20$ m/s from ground level.

$$h(t) = 20t - 4.9t^2$$

Find when the ball reaches maximum height: The ball is at maximum height when velocity equals zero:

$$v(t) = \mathcal{D}_t(20t - 4.9t^2) = 20 - 9.8t$$

$$0 = 20 - 9.8t$$

$$t = \frac{20}{9.8} \approx 2.04 \text{ seconds}$$

Maximum height:

$$h(2.04) = 20(2.04) - 4.9(2.04)^2 \approx 20.4 \text{ meters}$$

Key insight: Maximum (or minimum) values occur where the derivative equals zero!

3 Application: Economics

3.1 Cost Functions

Setup: Let $C(q)$ be the total cost of producing q units of a product.

Example:

$$C(q) = 5000 + 20q + 0.01q^2$$

- 5000: Fixed costs (rent, equipment)
- $20q$: Variable costs proportional to quantity
- $0.01q^2$: Additional costs from complexity at high volumes

3.2 Marginal Cost

Definition: The **marginal cost** is the derivative of the cost function:

$$MC(q) = \mathcal{D}_q C(q) = C'(q)$$

Interpretation: Marginal cost is approximately the cost of producing one additional unit.

Computing marginal cost:

$$MC(q) = \mathcal{D}_q(5000 + 20q + 0.01q^2) = 20 + 0.02q$$

At different production levels:

Quantity q	Total cost $C(q)$	Marginal cost $MC(q)$
100	$5000 + 2000 + 100 = 7100$	$20 + 2 = 22$
500	$5000 + 10000 + 2500 = 17500$	$20 + 10 = 30$
1000	$5000 + 20000 + 10000 = 35000$	$20 + 20 = 40$

Economic insight: Marginal cost increases with quantity—this reflects diminishing returns or capacity constraints.

3.3 Revenue and Profit

Revenue function: If each unit sells for price p :

$$R(q) = p \cdot q$$

Marginal revenue:

$$MR(q) = \mathcal{D}_q R(q)$$

Profit function:

$$\Pi(q) = R(q) - C(q)$$

Profit maximization: Maximum profit occurs where marginal revenue equals marginal cost:

$$MR(q) = MC(q)$$

Example: If $R(q) = 50q$ (price = \$50 per unit) and $C(q) = 5000 + 20q + 0.01q^2$:

$$MR(q) = 50$$

$$MC(q) = 20 + 0.02q$$

Setting $MR = MC$:

$$50 = 20 + 0.02q$$

$$30 = 0.02q$$

$$q = 1500 \text{ units}$$

4 Application: Biology

4.1 Population Growth

Exponential growth model: A population $P(t)$ growing without constraints follows:

$$P(t) = P_0 e^{rt}$$

where P_0 is the initial population and r is the growth rate.

Rate of population change:

$$\mathcal{D}_t P(t) = \mathcal{D}_t (P_0 e^{rt}) = P_0 \cdot r \cdot e^{rt} = r \cdot P(t)$$

Key result:

$$\boxed{\mathcal{D}_t P = rP}$$

The rate of change is proportional to the current population!

Example: A bacterial colony starts with $P_0 = 1000$ cells and doubles every hour.

Doubling time $T = 1$ hour means $P(1) = 2P_0$:

$$2P_0 = P_0 e^{r \cdot 1}$$

$$2 = e^r$$

$$r = \ln(2) \approx 0.693 \text{ per hour}$$

Population function:

$$P(t) = 1000 \cdot e^{0.693t} = 1000 \cdot 2^t$$

Growth rate at various times:

Time t (hours)	Population $P(t)$	Growth rate $\mathcal{D}_t P$
0	1,000	$0.693 \times 1000 = 693$ cells/hr
2	4,000	$0.693 \times 4000 = 2772$ cells/hr
5	32,000	$0.693 \times 32000 = 22,176$ cells/hr

4.2 Logistic Growth

More realistic model: Populations face resource limits. The logistic model:

$$P(t) = \frac{K}{1 + Ae^{-rt}}$$

where K is the carrying capacity.

Behavior:

- Initially grows exponentially (like before)
- Growth rate slows as P approaches K
- $\mathcal{D}_t P \rightarrow 0$ as $P \rightarrow K$

The derivative of logistic growth is more complex, but the key insight is: Maximum growth rate occurs at $P = K/2$ (half the carrying capacity).

5 Application: Environmental Science

5.1 Reservoir Dynamics

Setup: Let $V(t)$ represent the volume of water in a reservoir at time t (months).

Model with seasonal variation:

$$V(t) = 10000 + 2000 \sin\left(\frac{\pi t}{6}\right)$$

- Average volume: 10,000 acre-feet
- Seasonal fluctuation: ± 2000 acre-feet
- Period: 12 months

5.2 Rate of Change

Computing $\mathcal{D}_t V$:

$$\mathcal{D}_t V(t) = 2000 \cdot \frac{\pi}{6} \cos\left(\frac{\pi t}{6}\right) \approx 1047 \cos\left(\frac{\pi t}{6}\right)$$

Interpretation of the derivative:

- $\mathcal{D}_t V > 0$: Reservoir is filling (inflow $>$ outflow)
- $\mathcal{D}_t V < 0$: Reservoir is draining (outflow $>$ inflow)
- $\mathcal{D}_t V = 0$: Volume is momentarily constant (peak or trough)

At specific times:

Month t	Volume $V(t)$	Rate $\mathcal{D}_t V$	Interpretation
0 (Jan)	10,000	1047	Filling rapidly
3 (Apr)	12,000	0	Peak volume
6 (Jul)	10,000	-1047	Draining rapidly
9 (Oct)	8,000	0	Minimum volume
12 (Jan)	10,000	1047	Cycle repeats

5.3 Multi-Variable Example

Setup: A farmer's crop yield Y depends on water w (inches) and fertilizer f (pounds):

$$Y(w, f) = 50\sqrt{w} + 8f - 0.1f^2$$

Partial derivatives:

$$\begin{aligned}\mathcal{D}_w Y &= \frac{50}{2\sqrt{w}} = \frac{25}{\sqrt{w}} \\ \mathcal{D}_f Y &= 8 - 0.2f\end{aligned}$$

Interpretation:

- $\mathcal{D}_w Y = \frac{25}{\sqrt{w}}$: Yield gain per inch of water (decreases as w increases)
- $\mathcal{D}_f Y = 8 - 0.2f$: Yield gain per pound of fertilizer (decreases as f increases)

Optimal fertilizer: Set $\mathcal{D}_f Y = 0$:

$$\begin{aligned}8 - 0.2f &= 0 \\ f &= 40 \text{ pounds}\end{aligned}$$

Beyond 40 pounds, additional fertilizer actually *decreases* yield!

6 Summary

6.1 Key Concepts

1. **The $\mathcal{D}_t$ operator:** Takes a function and returns its rate of change with respect to t

2. **“With respect to”:** Specifies which variable changes while others are held fixed
3. **Physical meaning:**
 - $\mathcal{D}_t x$ = velocity (rate of change of position)
 - $\mathcal{D}_t v = \mathcal{D}_t^2 x$ = acceleration
4. **Economic meaning:**
 - $\mathcal{D}_q C$ = marginal cost
 - $\mathcal{D}_q R$ = marginal revenue
5. **Optimization:** Maxima and minima occur where $\mathcal{D}_t f = 0$

6.2 Derivatives of Common Functions

$f(t)$	$\mathcal{D}_t f(t)$
c (constant)	0
t^n	nt^{n-1}
e^{rt}	re^{rt}
$\sin(kt)$	$k \cos(kt)$
$\cos(kt)$	$-k \sin(kt)$
$\ln(t)$	$\frac{1}{t}$

6.3 Practice Exercises

Exercise 1 (Physics): A particle moves along a line with position $x(t) = t^3 - 12t + 5$.

- (a) Find the velocity function.
- (b) When is the particle at rest (velocity = 0)?
- (c) Find the acceleration function.
- (d) Is the particle speeding up or slowing down at $t = 1$?

Exercise 2 (Economics): A company's cost function is $C(q) = 800 + 15q + 0.02q^2$.

- (a) Find the marginal cost function.
- (b) What is the marginal cost at $q = 100$?
- (c) If the selling price is \$25 per unit, find the profit-maximizing quantity.

Exercise 3 (Biology): A population grows according to $P(t) = 500e^{0.05t}$.

- (a) What is the initial population?
- (b) Find the rate of population change as a function of time.
- (c) At what rate is the population growing when $t = 10$?

Exercise 4 (Environmental): A lake's water level follows $L(t) = 20 + 3 \sin\left(\frac{\pi t}{6}\right)$ meters, where t is in months.

- (a) Find $\mathcal{D}_t L(t)$.
- (b) When is the water level rising most rapidly?
- (c) When does the lake reach its maximum level?

Exercise 5 (Multi-variable): A company's profit depends on advertising a (thousands of dollars) and price p (dollars): $\Pi(a, p) = 100a^{0.5} + 50p - p^2 - 2a$.

- (a) Find $\mathcal{D}_a \Pi$ and $\mathcal{D}_p \Pi$.
- (b) What price maximizes profit (for a fixed advertising budget)?

Worked Solutions

Exercise 1 (Physics) — Solution: $x(t) = t^3 - 12t + 5$.

- (a) Velocity: $v(t) = \mathcal{D}_t x(t) = 3t^2 - 12$.
- (b) At rest when $v(t) = 0$: $3t^2 - 12 = 0 \Rightarrow t^2 = 4 \Rightarrow t = 2$ or $t = -2$.
- (c) Acceleration: $a(t) = \mathcal{D}_t v(t) = 6t$.
- (d) At $t = 1$: $v(1) = 3 - 12 = -9$ (moving left), $a(1) = 6 > 0$ (acceleration to the right). Velocity is negative and becoming less negative, so the particle is **slowing down** (speed decreasing).

□

Exercise 2 (Economics) — Solution: $C(q) = 800 + 15q + 0.02q^2$.

- (a) Marginal cost: $\mathcal{D}_q C(q) = 15 + 0.04q$.
- (b) At $q = 100$: marginal cost $= 15 + 4 = 19$ (dollars per unit).
- (c) Revenue $R(q) = 25q$, profit $\Pi(q) = R(q) - C(q) = 25q - 800 - 15q - 0.02q^2 = 10q - 800 - 0.02q^2$. Set $\mathcal{D}_q \Pi = 10 - 0.04q = 0 \Rightarrow q = 250$. Profit-maximizing quantity is $q = 250$ units.

□

Exercise 3 (Biology) — Solution: $P(t) = 500e^{0.05t}$.

- (a) Initial population: $P(0) = 500e^0 = 500$.
- (b) Rate of change: $\mathcal{D}_t P(t) = 500 \cdot 0.05 \cdot e^{0.05t} = 25e^{0.05t}$.
- (c) At $t = 10$: rate $= 25e^{0.5} \approx 25 \times 1.649 \approx 41.2$ (units per time).

□

Exercise 4 (Environmental) — Solution: $L(t) = 20 + 3 \sin\left(\frac{\pi t}{6}\right)$.

- (a) $\mathcal{D}_t L(t) = 3 \cdot \frac{\pi}{6} \cos\left(\frac{\pi t}{6}\right) = \frac{\pi}{2} \cos\left(\frac{\pi t}{6}\right)$.
- (b) Water level rises most rapidly when $\cos\left(\frac{\pi t}{6}\right)$ is maximum (i.e., $= 1$). So $\frac{\pi t}{6} = 0, 2\pi, 4\pi, \dots$, hence $t = 0, 12, 24, \dots$ months. At $t = 0$ (and every 12 months), the level is rising most rapidly.
- (c) Maximum level when $\sin\left(\frac{\pi t}{6}\right) = 1$, i.e., $\frac{\pi t}{6} = \frac{\pi}{2}, \frac{5\pi}{2}, \dots$, so $t = 3, 15, 27, \dots$ months.

□

Exercise 5 (Multi-variable) — Solution: $\Pi(a, p) = 100a^{0.5} + 50p - p^2 - 2a$.

(a) $\mathcal{D}_a\Pi = 100 \cdot 0.5a^{-0.5} - 2 = \frac{50}{\sqrt{a}} - 2$; $\mathcal{D}_p\Pi = 50 - 2p$.

(b) For fixed a , maximize in p : set $\mathcal{D}_p\Pi = 50 - 2p = 0 \Rightarrow p = 25$ dollars. Second derivative in p is $-2 < 0$, so $p = 25$ maximizes profit (for fixed a).

□